HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CEMENT & BUILDING MATERIALS · PROCUREMENT & FINANCE

TEN PLANTS · FUEL AND RAW MATERIALS · ONE COST MODEL

Procurement Cost Intelligence

Fuel and raw materials are a cement plant's biggest controllable costs, and their prices swing every month. HMC put every tonne, every lane and every rupee of material cost on one screen shared by the buyer and the controller.

6 → 1 weeks after month end to explain 90% of cost variance

10 plants
Plants on one supply map
integrated and grinding, east and north

34 lanes
Inbound lanes costed
rail, road and sea, to the plant gate

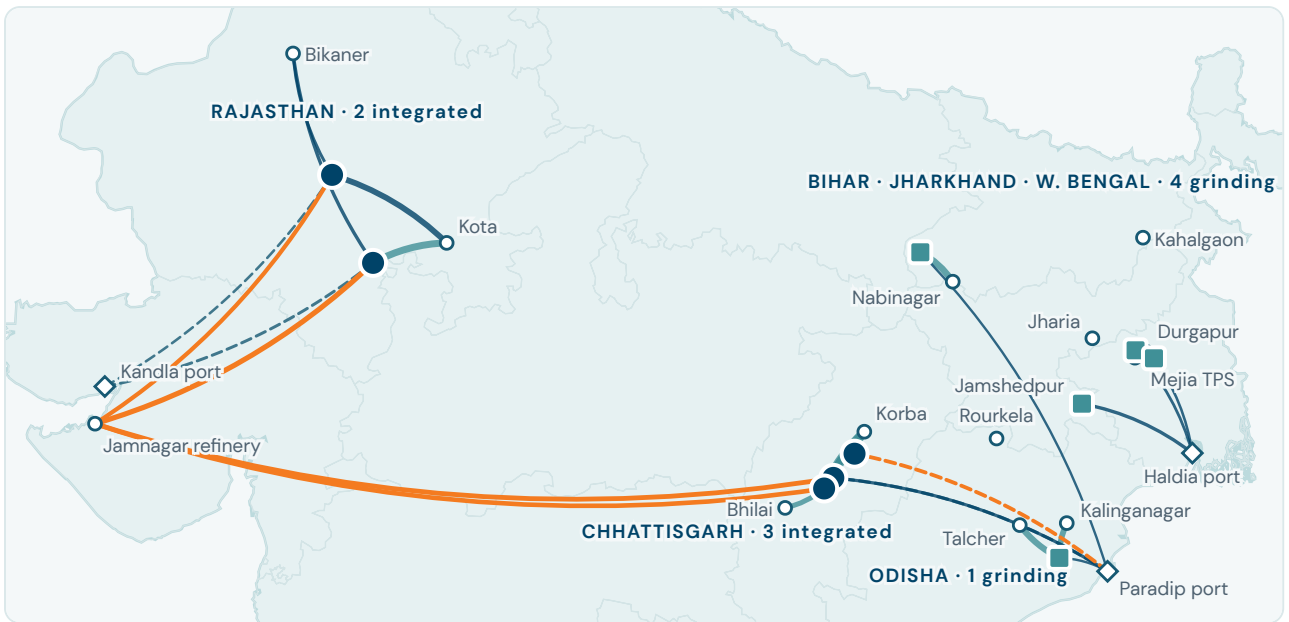
6 inputs
Material and fuel inputs tracked
limestone to gypsum

3.4%
Lower material and fuel cost per tonne
fuel mix, sourcing, contracts

Rings: 5 integrated plants (white) and 5 grinding units; lanes by mode (rail 16, road 14, sea + rail 4); one per input; ₹2,210/t baseline by category, saving in orange. Figures indicative.

01 Every inbound lane on one map

Supply network map · plants, sources and the 34 costed lanes



Line width = tonnes moved

Plant locations approximate; sources, lanes and volumes synthetic. Captive limestone and internal clinker transfers not drawn. Boundaries: Survey of India outline via DataMeet (CC BY).

BIGGEST SINGLE INPUT

Petcoke comes from the coast

The largest line in kiln cost (₹470/t of cement before) reaches inland plants by rail from a west-coast refinery and by sea and rail through eastern ports.

COMPLEX NETWORK

17 sources, three modes

Fly ash, slag, gypsum, coal and petcoke reach each plant from coalfields, thermal plants, steel plants, refineries and ports.

- **Thermal outage**: two West Bengal units lose local fly ash; reroute to Kahalgaon
- **Port congestion**: Paradip imports shift to Haldia
- **Slag shortfall**: slag drawn from Rourkela instead of Jamshedpur
- **Refinery shutdown**: petcoke imported via Kandla and Paradip

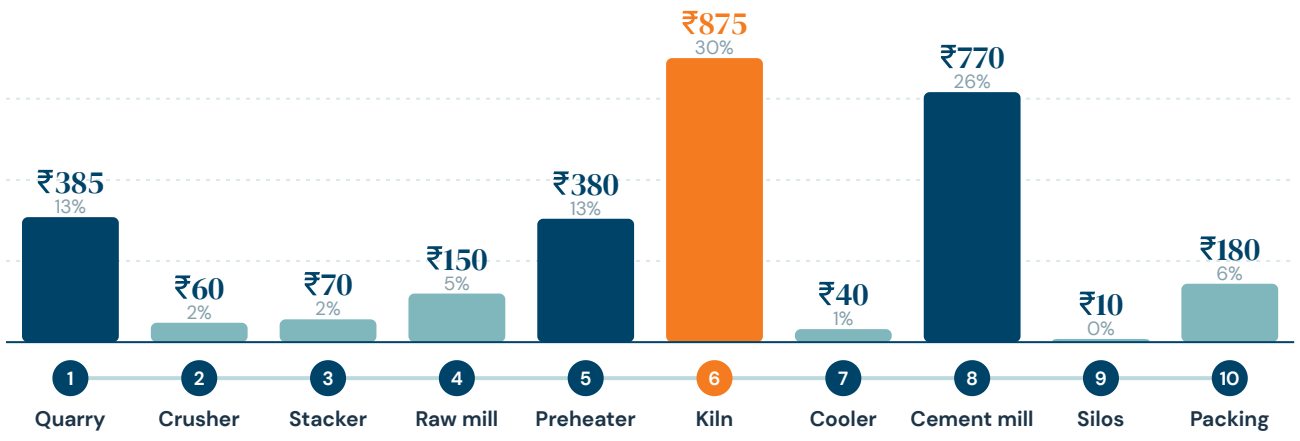
WHAT CHANGED

Plants become nodes, not files

Disruptions are switched on and the reroute is costed before it happens; new plants are added as nodes on the same model.

02 Where the cost enters the process

Cost per tonne of cement by stage · schematic plant, figures indicative



Quarry to dispatch · ₹ per tonne of cement, material, fuel and power

STAGE 6 · ROTARY KILN, ABOUT 1,450°C

Where procurement matters most

Most fuel is burnt here: ₹875 of the ₹2,920 per tonne plant-gate material, fuel and power cost.

Petcoke	₹470
Imported coal	₹260
Domestic coal	₹110
Refractories	₹35

Lever: the petcoke–coal mix within sulphur limits, bought on landed ₹ per million kcal, not ₹ per tonne.

Stage 8, the cement mill (₹770): fly ash and slag landed cost decide which blend is cheapest to make.

03 Four pain points, four fixes

Before → after

BEFORE
Price swings
Petcoke and imported coal move with global markets and the rupee; risk was reacted to after the bill.

PRICE VOLATILITY LAB

AFTER
Shocks simulated first
Price shocks run before buying; the fuel mix is optimised within kiln limits at current prices.

BEFORE
Complex supply network
Supplier lists per plant; many sources by rail, road and sea.

SUPPLY NETWORK MAP

AFTER
One live map
Every lane on one map, with disruption scenarios and their cost.

BEFORE
Landed cost in pieces
Base price compared first, freight, handling and quality added later from other files.

ONE LANDED-COST RULE

AFTER
One number
Price, freight, handling and quality in a single landed cost per supplier.

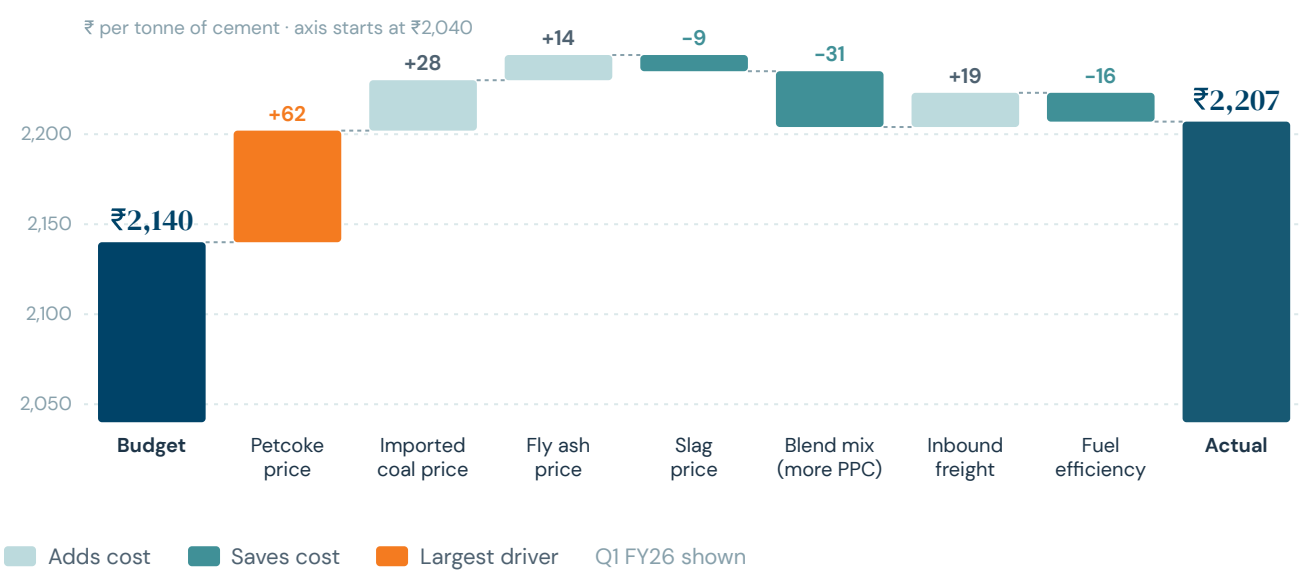
BEFORE
Variance explained late
Finance learned why cost per tonne missed budget weeks after close.

FINANCE DESK

AFTER
Bridge in week one
A variance bridge by driver in the first week; stock cover set against cash and stock-out risk.

04 Budget to actual, driver by driver

Finance desk variance bridge · material and fuel cost per tonne · synthetic data



SAME BRIDGE, EVERY QUARTER · ₹ PER TONNE

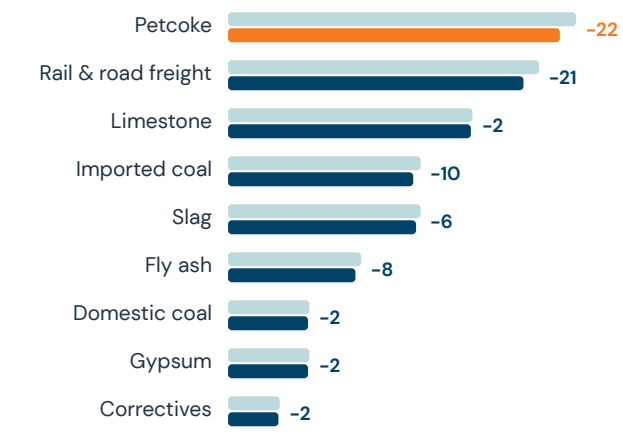
QUARTER	BUDGET	ACTUAL	GAP	LARGEST DRIVER
Q1 FY26	₹2,140	₹2,207	+67	Petcoke +62
Q2 FY26	₹2,140	₹2,159	+19	Inbound freight +26
Q3 FY26	₹2,140	₹2,056	−84	Petcoke −38
Q4 FY26	₹2,140	₹2,163	+23	Petcoke +41

ONE LANDED-COST DEFINITION

Landed cost = price + freight + handling + levies, adjusted for moisture or calorific value. Variance is split into price, volume, mix, freight and efficiency.

05 Cost per tonne, by input

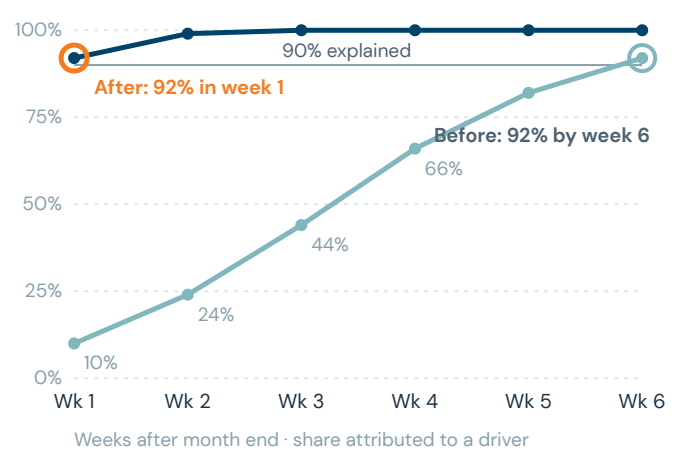
₹ per tonne of cement, before (light) and after (navy) · indicative



₹2,210 → ₹2,135 per tonne, −₹75 (−3.4%)

06 Variance, explained sooner

Share of monthly variance attributed to a driver · indicative



07 How HMC ran it

2 weeks — **Discover · follow the tonne**
Every input traced from source to kiln; supply network inventory and cost tree.

2 weeks — **Design · one definition**
Landed-cost rules, cost per tonne by stage and a variance bridge, agreed once.

5 weeks — **Build · map, model, lab, desk**
Four tools on one data model: map, price lab and fuel-mix optimiser, finance desk.

2 cycles — **Adopt · monthly rhythm**
Weekly buying review on the price lab; month-end variance bridge for finance.

∞ ongoing — **Scale · more inputs, plants**
Power and packing next; forecasts feed the annual budget.

What any manufacturer can borrow

01 Compare landed, not quoted
The cheapest price at the source is often not the cheapest tonne at the plant gate.

02 Simulate before you buy
Knowing what a price shock does to a bag of cement turns market news into a buying decision.

03 One model for buyer and controller
When procurement and finance share the cost tree, variance stops being a debate and becomes an action list.

Built with **Power Query** · **Python** · **Power BI** · **Excel models** · **GIS** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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